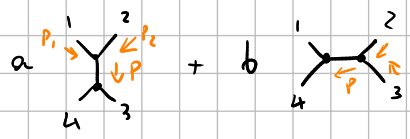
4 point

$B =$ 

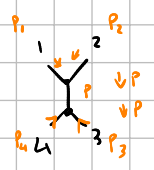
$$= \frac{a}{X_{13}} + \frac{b}{X_{24}}$$

X_{13}, X_{24} independent planar variables

$X_{ij} = (P_i + P_{i+1} + \dots + P_{j-1})^2$

$X_{ii} = X_{i\,i+1} = 0$

$X_{in} = 0$



each vertex $\propto \delta$ of incoming − outgoing momentum

each internal line $= \int dp \, \frac{1}{p^2}$ of associated momentum

$$\int dp \; \delta(P_1 + P_2 - P) \; \overbrace{\delta(P_3 + P_4 + P) \, \frac{1}{p^2}}^{f(p)} = \frac{\delta(P_1 + P_2 + P_3 + P_4)}{(P_1 + P_2)^2} = \frac{\delta(P_1 + P_2 + P_3 + P_4)}{X_{13}}$$

$$\int dp \; \delta(p - q) \, f(p) = f(q)$$

---

then impose 1-zero

require that $c_{13} = 0 \Rightarrow B = 0 \;\; \forall X$

$c_{13} = 0$ $(n = 4)$

$\parallel$

$X_{13} + X_{24} - \cancel{X_{14}}^{\,0} - \cancel{X_{34}}^{\,0}$

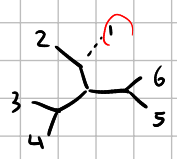
$c_{13} = c_{14} = \dots = c_{1\,n-1} = 0$

$(n-3)$ constraints

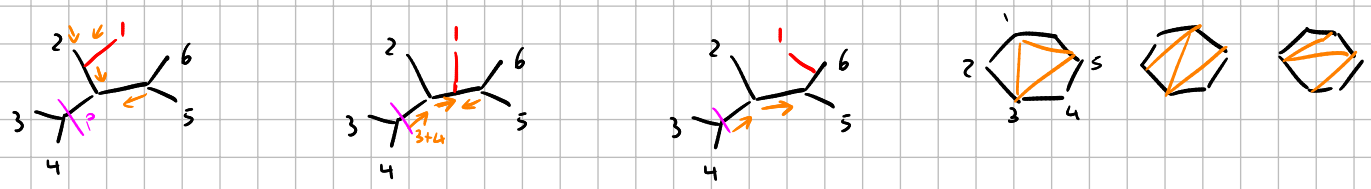
$$B = \frac{a}{X_{13}} + \frac{b}{X_{24}} = \frac{a - b}{X_{13}} \qquad \text{so} \quad a = b$$

---

ex: $n = 6$ choose a 6-leg diagram with leg 1 removed

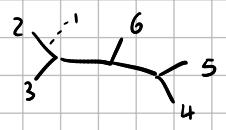
start with 

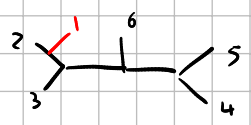
add leg 1 back in in all possible ways (according to order)

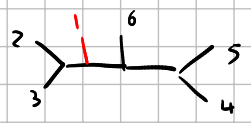


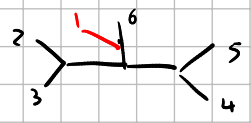
$$S_1 = \frac{a_1}{X_{13}\, X_{51}\, X_{35}} + \frac{a_2}{X_{25}\, X_{15}\, X_{35}} + \frac{a_3}{X_{25}\, X_{26}\, X_{35}}$$

another D-subset





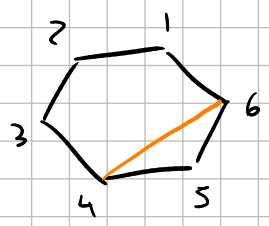


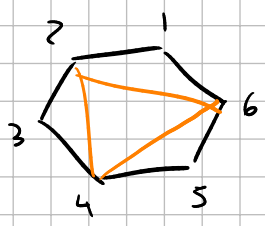


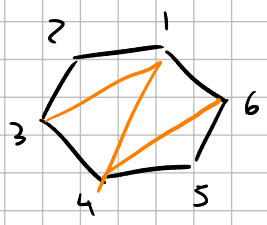
$$S_2 = \frac{a_4}{X_{13}\, X_{46}\, X_{41}} + \frac{a_5}{X_{24}\, X_{46}\, X_{41}} + \frac{a_6}{X_{24}\, X_{46}\, X_{26}}$$

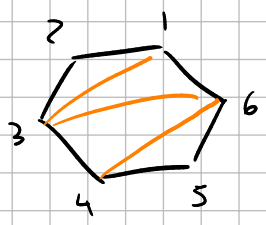
$S_1 + S_2$ all the diagrams that contain X_{46} and X_{35}

<u>all diagrams</u> that have $X_{46} = (P_4 + P_5)^2$









look at $S_1 + S_2$ and see that it satisfies 1-zero ($c_{13} = c_{14} = c_{15} = 0$)

$\Rightarrow X_{2j} = X_{1j} - X_{13} \quad j = 4, \dots, 6$

if $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 + S_2 = 0 \quad \forall X_{ij}$ subject to constraint

<u>then</u> $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 = 0$ and $S_2 = 0 \quad \forall X_{ij}$

$$S_1 + S_2 = \frac{a_1}{X_{13}\, X_{51}\, X_{35}} + \frac{a_2}{X_{25}\, X_{15}\, X_{35}} + \frac{a_3}{X_{25}\, X_{26}\, X_{35}}$$

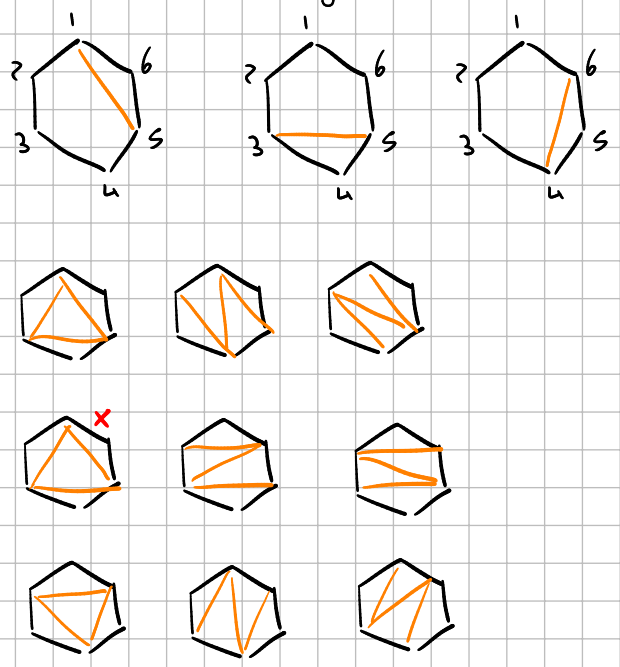
$$+ \frac{a_4}{X_{13}\, \boxed{X_{46}}\, X_{41}} + \frac{a_5}{X_{24}\, \boxed{X_{46}}\, X_{41}} + \frac{a_6}{X_{24}\, \boxed{X_{46}}\, X_{26}}$$

$$= \frac{1}{X_{35}} \underbrace{(\,-\,-\,)}_{=0} + \frac{1}{X_{46}} \underbrace{(\,-\,-\,)}_{=0}$$

$X_{2j} = X_{1j} - X_{13}$

all possible diagrams with 6 legs

group them by



splitting depending on z scaling of propagators

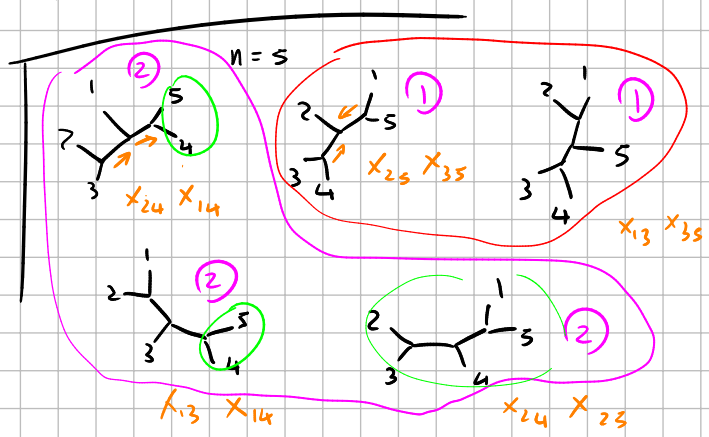
<u>boundary propagators</u>: the ones that scale linearly with z under BCFW shift (X_{1j} and X_{2j})

group by # of boundary propagators

→ group them by scaling under BCFW shift

$$B = \boxed{B_0} + \boxed{B_{-1}} + \boxed{B_{-2}} + \dots$$

0 boundary propagators — 1 bound. prop. — 2 boundary prop.



<u>bound. prop.</u>: X_{13} X_{14}
X_{24} X_{25}